📊 Interactive Sales Forecast Dashboard

A **Streamlit web application** for sales forecasting using **ARIMA** and **Prophet** models. Designed to provide **executive-ready insights**, interactive visualizations, and actionable KPIs for sales planning and inventory management.

🏛️ Background

Accurate sales forecasting is crucial for businesses to optimize **inventory management, staffing, and promotional strategies**. Traditional forecasting methods often fail to adapt to trends and seasonal variations, leading to overstock or stockouts, and ultimately, loss of revenue.

This project provides a modern, **data-driven solution** that leverages **ARIMA** and **Prophet** models to generate reliable forecasts while presenting results in an **executive-friendly dashboard**.

⚠️ Problem Statement

Businesses often face challenges such as:

- Poor demand predictions leading to **overstock or stockouts**.
- Difficulty interpreting complex statistical outputs for strategic decisions.
- Lack of a unified dashboard combining forecasts, KPIs, and residual analysis.
- Time-consuming manual analysis for executives needing **quick insights**.

🎯 Objectives

The main objectives of this project are:

1. **Develop a user-friendly dashboard** for sales forecasting.
2. Enable **forecasting using ARIMA and Prophet models**.
3. Provide **interactive visualizations** for actual vs forecasted sales and residuals.
4. Calculate **key performance indicators** (MAPE, MAE, RMSE) with **color-coded executive interpretation**.
5. Allow users to **upload their own sales data** or use a demo dataset.
6. Facilitate **data export** for further analysis or reporting.

🛠️ Methodology

1. **Data Preparation**

   Source of Data:

- Users can upload their own historical sales data in CSV format with `date` and `sales` columns.
- The app also includes a **demo dataset** generated programmatically for demonstration purposes.
- Real-world deployment should use verified historical sales records from your business's ERP, CRM, or POS systems

    - Upload CSV with `date` and `sales` columns, or use built-in demo data.
    - Normalize column names and ensure datetime formatting.
2. **Train-Test Split**
    - Split data into 80% training and 20% testing for model evaluation.
3. **Modeling**
    - **ARIMA**: Uses past sales trends to forecast future demand.
    - **Prophet**: Handles seasonality, trends, and holidays for robust forecasting.
4. **Evaluation Metrics**
    - **MAPE**: Mean Absolute Percentage Error
    - **MAE**: Mean Absolute Error
    - **RMSE**: Root Mean Squared Error
5. **Visualization & Dashboard**
    - Forecast vs Actual interactive Plotly chart
    - Residuals visualization
    - KPI display with **green/orange/red interpretation**
    - CSV download for forecasted data

📊 Results / Conclusions

- The dashboard provides **highly interpretable insights** for executives.
- Users can identify trends, forecast accuracy, and deviations efficiently.
- MAPE values indicate forecast reliability:
    - **Green (<5%)**: Excellent forecast accuracy
    - **Orange (5–15%)**: Moderate accuracy
    - **Red (>15%)**: Low accuracy, signals potential model improvement
- Residual analysis helps detect systematic errors or anomalies.

💡 Recommendations

1. **Use ARIMA for short-term forecasts** with stable historical trends.
2. **Use Prophet for longer-term forecasts** or datasets with seasonality.
3. **Regularly update data** and retrain models to maintain forecast accuracy.
4. **Leverage KPI and residual charts** to guide inventory, staffing, and marketing strategies.
5. **Integrate with business dashboards** for executive decision-making and automated reporting.

   sales-forecast-dashboard/

```
│
├── app.py            # Streamlit dashboard
├── model.py          # Forecasting functions (ARIMA & Prophet)
├── demo_sales.csv    # Sample sales CSV
├── requirements.txt  # Python dependencies
└── README.md         # Project documentation
```

Dependencies

- streamlit
- pandas
- numpy
- plotly
- statsmodels
- prophet
- scikit-learn

How to use the App

- Run The Streamlit App
- Upload your own CSV with `date` and `sales` columns, or use the demo data.
- Choose a **model** (ARIMA or Prophet) and **forecast horizon**.
- Explore **forecast plots, residuals, and KPI metrics**.
- Download the forecasted data as a CSV.

👨‍💻 About the Developer

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